

Mathematical Methods 2 (Group Theory) Notes

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Lecture 01

- **Why Group Theory?**

Groups are relevant whenever there is a symmetry of a physical system

- **Axioms:**

A group is a set of elements $G = \{g_i\}$ together with a map¹ $G \times G \rightarrow G$ with $g_i \circ g_j \in G$ ² and satisfying following

1. **Associativity:** $g_i \in G \forall i (g_1 g_2) g_3 = g_1 (g_2 g_3)$
2. **Identity:** There is an element e which satisfies

$$eg = ge \quad \forall g \in G$$

3. **Inverse:** For every $g \in G$, there is an inverse element g^{-1} , such that

$$gg^{-1} = g^{-1}g = e$$

- **Properties of Group:**

1. In general, $g_1 g_2 \neq g_2 g_1$. If $g_1 g_2 = g_2 g_1 \quad \forall g_1 g_2 \in G$, then the group is called *Commutative* or *Abelian*
2. **Order of Group**³
The order of the group is the number of elements in the group. It is denoted by $|G|$
3. **Order of a Group Element**
order $g = p, g^p = e$
4. **For Continuous Groups:**
Dimension: The number of parameters you need to describe a

¹It sometimes also referred as *group product* or *operation*

²This is called *closure* property

³It is defined only for finite discrete groups

group.

Compactness: Compactness refers to whether the range of parameters is *compact*⁴

Example:

$$U(1) = e^{i\theta}; 0 \leq \theta \leq 2\pi$$

is compact as parameter θ is closed and bounded⁵

Example: SO(3): 3D Rotation: 2-D Rotation

$$R(\theta) = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$-1 \leq \cos \theta, \sin \theta \leq 1$$

Example: SO(1,3) $SO(1,3) = \{\Lambda | \Lambda^T \eta \Lambda = \Lambda, \det \Lambda = 1\}$

$$\eta = \text{diag}(-1, 1, 1, 1)$$

for a boost in x-direction

$$\Lambda = \begin{pmatrix} \cosh \eta & \sinh \eta & 0 & 0 \\ \sinh \eta & \cosh \eta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

This is not compact as the transformation matrix is closed but not bounded

- **Finite Discrete Groups**

The n -th root of unity form an Abelian group of order n under multiplication

$$\theta = e^{2\pi i/n}; \quad G = \{\theta^p, p = 1, 2, \dots, n\}$$

Identity is $\theta^n = 1$, Inverse: $\theta^{-p} = \theta^{n-p}$

$\theta \rightarrow$ generating element

$\theta^n = e$ is called the cyclic group of order $n : \mathbb{Z}_n$

- **Symmetric group S_n**

permutation of n things

- **Infinite discrete groups**

$(\mathbb{Z}, +)$: set of Integers under addition

⁴closed and bounded

⁵the bounds are included

- **Continuous Compact groups:**

1. Orthogonal groups $O(n) \implies O^T O = \mathbb{I}$
2. Unitary groups: $U(n) \implies U^\dagger U = \mathbb{I}$

- Example of non-compact set $SO(m, n) \quad m, n \in \mathbb{Z}, E(n)$ ⁶

- **Subgroup:**

A subgroup H of G is a subset of elements which forms a group under product of G .

A proper subgroup of H is when $H \neq G$ and is denoted by $H < G$

- **Properties of Subgroup:**

1. Every subgroup must contain identity
 2. A subset $H \subset G$ is a subgroup $\forall h_1, h_2 \in H$ then $h_1 h_2^{-1} \in H$
 3. $\{e\}$ and G are trivially subgroups
- $Z(G) = \{ h \in G | hg = gh \quad \forall g \in G \}$
 $Z(G)$ is an Abelian subgroup

⁶It is like rotation + translation in Euclidean space with operation $\mathbf{x}' = O\mathbf{x} + \mathbf{a}$